

1. **Secure Dark Web Investigation Environment** A cyber security analyst receives information about a suspicious onion website spreading malware. Before accessing the site, the analyst must create a secure isolated virtual machine environment to avoid malware infection and identity leakage. Tools: VirtualBox, Whonix, Tor Browser

Step 1 — Install VirtualBox

- Download & install VirtualBox
- Open VirtualBox

Step 2 — Download Whonix

Download:

- Whonix Gateway
- Whonix Workstation

Step 3 — Import Whonix

- VirtualBox → File → Import Appliance
- Import Gateway & Workstation

Step 4 — Start Gateway

- Start Whonix-Gateway
- Wait for:

Tor is ready

Step 5 — Start Workstation

- Start Whonix-Workstation

Step 6 — Open Tor Browser

Inside Workstation:

- Open Tor Browser

Step 7 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

2. **Employee Credential Leak Investigation** A multinational company notices unauthorized logins in employee accounts. Students must investigate whether company email credentials are leaked on dark web forums or breach databases. Tools: IntelligenceX, Have I Been Pwned, Tor Browser

1. **Open Tor Browser**

2. **Open:**

<https://haveibeenpwned.com>

3. **Enter sample email:**

employee@company.com

Click:

pwned?

4. **Open:**

<https://intelx.io>

5. **Search:**

company.com

OR

[employee@company.com](#)

3. **Malware Payload Behavior Analysis** A ransomware sample uploaded on a darknet forum is suspected of targeting educational institutions. Students must safely analyze the malware behavior in a sandbox environment. Tools: REMnux, Metasploit Framework, VirusTotal

Step 1 — Start REMnux VM

- Open VirtualBox
- Start REMnux VM

Step 2 — Open Terminal

Inside REMnux:

- Open terminal

Step 3 — Navigate to Malware Sample

Command:

```
cd ~/Desktop
```

Check sample:

```
ls
```

Example:

```
sample.exe
```

Step 4 — Generate SHA256 Hash

Command:

```
sha256sum sample.exe
```

Output:

```
8f3d2ab34a....
```

Step 5 — Open VirusTotal

Open:

```
https://www.virustotal.com
```

Paste generated hash into search.

Step 6 — Analyze Detection Results

Observe:

- Detection ratio
- Malware family
- Threat indicators

Step 7 — Open Metasploit

Back to terminal:

```
msfconsole
```

Wait for framework to load.

4. Hidden Service Structure Mapping A cybercrime investigation unit discovers a suspicious onion service linked to illegal trading. Students must identify linked pages, login portals, and communication channels without interacting with illegal content. Tools: OnionScan, Recon-ng, Tor Browser

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Observe:

- Homepage
- Search options
- Linked pages

Step 4 — Open Terminal

Run OnionScan:

```
onionscan duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion
```

Step 5 — Open Recon-ng

Command:

```
recon-ng
```

Step 6 — Analyze Structure

Identify:

- Linked pages
- Login portals
- Contact/communication pages

DO NOT interact.

5. Darknet Marketplace Vendor Assessment Law enforcement suspects that fake sellers are scamming users in a darknet marketplace. Students must analyze vendor ratings, product categories, and payment methods to identify fraud indicators. Tools: Tor Browser, Maltego CE, Screenshot Tools

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Step 4 — Analyze Vendor Indicators

Check:

- Ratings/reviews
- Product categories
- Payment methods
- Suspicious patterns

Step 5 — Open Maltego CE

Start:

Maltego CE

Step 6 — Create Graph

Add:

- Vendor
- Marketplace
- Payment wallet
- Communication method

Output

- Vendor assessment performed
- Fraud indicators identified
- Relationships visualized

6. Ransomware Bitcoin Wallet Investigation A hospital affected by ransomware provides a Bitcoin wallet used for ransom payment. Students must trace the wallet activity and identify suspicious transaction patterns. Tools: Blockchain Explorer, BTC Explorer, Chainalysis Reactor (Demo)

Step 1 — Open Blockchain Explorer

Open:

<https://www.blockchain.com/explorer>

Step 2 — Search Bitcoin Wallet

Use sample wallet:

1BoatSLRHtKNngkdXEeobR76b53LETtpyT

Paste into search bar.

Step 3 — Analyze Wallet Information

Step 4 — Scroll to Transaction History

Observe:

- Incoming/outgoing transactions
- Transaction IDs
- Dates & amounts

Step 5 — Open Blockchair Explorer

Open:

<https://blockchair.com/bitcoin>

Search same wallet.

Observe:

- Additional transaction analytics
- Linked transactions
- Address behavior

Step 6 — Open Chainalysis Reactor (Demo)

Open demo version if available.

Observe:

- Transaction flow graph
- Wallet relationships
- Suspicious transfer patterns

Step 7 — Identify Suspicious Patterns

- Zero final balance
- Unknown recipient wallets

6. Planned Cyber Attack Intelligence Gathering A threat intelligence team receives alerts about a possible cyber attack being discussed on underground forums. Students must collect and correlate intelligence from surface web and dark web sources. Tools: IntelligenceX, SpiderFoot, Maltego CE

tep 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

sakec.ac.in

Observe:

- Text files
- Database leaks
- Related intelligence

Step 2 — Start SpiderFoot

Open terminal and type:

```
spiderfoot -l 127.0.0.1:5001
```

Press Enter.

Step 3 — Open SpiderFoot Dashboard

Open browser:

<http://127.0.0.1:5001>

Step 4 — Create New Scan

Scan Name:

ThreatScan

Scan Target:

sakec.ac.in

Keep:

All

selected.

Click:

Run Scan

Step 5 — Open Maltego

Open terminal:

```
maltego
```

Wait for workspace to open.

7. Vulnerability Exploitation Validation A company security team wants to verify whether an outdated server is vulnerable to known exploits. Students must perform ethical exploit verification in an isolated lab setup.
Tools: Metasploit Framework, Kali Linux, VirtualBox

Step 1 — Start Virtual Machines

Open:

VirtualBox

Start:

- Kali Linux VM
- Vulnerable target VM (example: Metasploitable)

Step 2 — Check Target IP

In target VM terminal:

ifconfig

OR

ip a

Note target IP.

Example:

192.168.56.101

Step 3 — Open Metasploit

In Kali terminal:

msfconsole

Wait for:

msf6 >

Step 4 — Search Vulnerability Exploit

Command:

search vsftpd

Observe exploit results.

Step 5 — Select Exploit

Command:

use exploit/unix/ftp/vsftpd_234_backdoor

Step 6 — Set Target IP

Command:

set RHOSTS 192.168.56.101

Step 7 — Run Exploit Verification

Command:

run

Observe:

- Vulnerability verification
- Session opened
- Exploit success/failure

 SS4 → Exploit verification result

8. Banking Phishing Campaign Investigation A phishing URL targeting online banking users is discovered on a darknet forum. Students must analyze the hosting details, related domains, and possible attacker infrastructure. Tools: URLScan, IntelligenceX, WHOIS Lookup

Step 1 — Open URLScan

Open:

<https://urlscan.io/>

Step 2 — Analyze Suspicious URL

Search sample phishing URL:

<http://fakebank-login.com>

Observe:

- Hosting IP
- Server details
- Linked resources
- Domain behavior

Step 3 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

fakebank-login.com

Observe:

- Related domains
- Breach references
- Infrastructure indicators

 SS2 → IntelligenceX search results

Step 4 — Open WHOIS Lookup

Open:

<https://whois.domaintools.com/>

Search:

fakebank-login.com

Observe:

- Registrar
- Creation date
- Name servers
- Registrant details

 SS3 → WHOIS lookup result

9. Dark Web User Communication Analysis Investigators suspect a group of cybercriminals are coordinating through darknet discussion forums. Students must analyze posting behavior, usernames, and communication patterns. Tools: Tor Browser, Maltego CE, Spreadsheet Tools

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open Safe Onion Search Engine

Search/open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Step 4 — Search Sample Keywords

Search:

cyber forum

OR

hacking discussion

Observe:

- Usernames
- Post timings
- Discussion topics
- Communication style

DO NOT interact/login.

 SS2 → Search/forum results

Step 5 — Open Spreadsheet Tool

Open:

- Excel
OR
- Google Sheets

Create columns:

| Username | Post Time | Topic | Communication Pattern |

Step 6 — Enter Sample Observations

Example:

Username	Post Time	Topic	Communication Pattern
darkuser01	10:30 PM	Malware	Frequent posting
cyberx99	11:15 PM	Phishing	Anonymous communication
shadowsec	12:05 AM	Data Leak	Repeated activity

 SS3 → Spreadsheet analysis table

Step 7 — Open Maltego CE

Open terminal:

maltego

Wait for workspace.

 SS4 → Maltego workspace

Step 8 — Create Relationship Graph

Add entities:

- Usernames
- Forums
- Topics
- Communication channels

Observe relationships.

10. University Data Breach Investigation A university receives reports that student records are available for sale online. Students must search for leaked information and prepare a basic risk assessment report. Tools: IntelligenceX, DeHashed, OSINT Framework

Step 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

university.edu

OR sample:

student@university.edu

Observe:

- Leaked records
- Breach references
- Public exposure

 SS1 → IntelligenceX search results

Step 2 — Open DeHashed

Open:

<https://dehashed.com/>

Search:

university.edu

Observe:

- Exposed emails
- Password leaks
- Breach data

 SS2 → DeHashed search results

Step 3 — Open OSINT Framework

Open:

<https://osintframework.com/>

Observe:

- Investigation categories
- Email investigation tools
- Breach analysis resources

 SS3 → OSINT Framework page

Step 4 — Analyze Risk Indicators

Identify:

- Exposed student emails
 - Password exposure
 - Public breach references
 - Risk severity
-

Step 5 — Prepare Basic Risk Assessment

Write observations:

- Possible data exposure
- Risk to students
- Recommended actions

Example:

Enable MFA, reset passwords, monitor suspicious access

Output

- Breach information identified
- Risk indicators analyzed
- Basic risk assessment prepared

11. Threat Actor Infrastructure Investigation A cyber threat group is suspected of operating multiple phishing domains and servers. Students must map the infrastructure connections between domains, IP addresses, and hosting providers. Tools: Shodan, Recon-ng, Maltego CE

Step 1 — Open Shodan

Open:

<https://www.shodan.io/>

Search sample:

apache

OR

nginx

Observe:

- IP addresses
- Hosting providers
- Open services
- Server details

 SS1 → Shodan search results

Step 2 — Open Recon-ng

Open terminal:

recon-ng

Wait for:

[recon-ng][default] >

 SS2 → Recon-ng console

Step 3 — Use Marketplace Search (Optional)

Inside Recon-ng:

marketplace search

Observe available modules.

Step 4 — Open Maltego CE

Open terminal:

maltego

Wait for workspace.

 SS3 → Maltego workspace

Step 5 — Create Infrastructure Graph

Add entities:

- Domains
- IP addresses
- Hosting providers
- Servers

Analyze:

- Relationships
- Connected infrastructure
- Suspicious hosting patterns

13. Ethical Remote Exploitation Demonstration

Students are asked to demonstrate exploitation of a vulnerable web application in a controlled laboratory environment for awareness training.

Tools: Metasploit Framework, DVWA, Kali Linux

Start Apache & MySQL Services

Open terminal in Kali.

Run:

```
sudo service apache2 start
```

Then:

```
sudo service mysql start
```

Check Apache Status

Run:

```
sudo service apache2 status
```

You should see:

```
active (running)
```

SS1

Take screenshot:

- Apache running status

Caption:

Figure 1: Apache Web Server Running

Install DVWA (If Not Installed)

Go to:

```
cd /var/www/html
```

Clone DVWA:

```
sudo git clone https://github.com/digininja/DVWA.git
```

Configure Permissions

Run:

```
sudo chmod -R 777 /var/www/html/DVWA
```

Create DVWA Database

Open browser:

```
http://127.0.0.1/DVWA/setup.php
```

Click:

Create / Reset Database

SS2

Take screenshot:

- DVWA setup page

Caption:

Figure 2: DVWA Database Setup

Open DVWA Login Page

Open:

```
http://127.0.0.1/DVWA/login.php
```

Login:

Username: admin

Password: password

SS3

Take screenshot:

- DVWA dashboard/login successful

Caption:

Figure 3: DVWA Vulnerable Web Application Dashboard

Set Security Level LOW

Inside DVWA:

- Click:

DVWA Security

Select:

Low

Click:

Submit

SS4

Take screenshot:

- Security level LOW

Caption:

Figure 4: DVWA Security Level Configuration

Open Metasploit

Back to Kali terminal:

msfconsole

Wait for:

msf6 >

SS5

Take screenshot:

- Metasploit console

Caption:

Figure 5: Metasploit Framework Running

Search Exploit

Run:

search dvwa

Observe available modules.

1 1 Use Exploit Module

Example:

use exploit/unix/webapp/php_include

1 2 Set Target IP

Run:

set RHOSTS 127.0.0.1

1 3 Run Exploit

Run:

run

Observe:

- Vulnerability demonstration
 - Session output
-

SS6

Take screenshot:

- Exploit execution/result

Caption:

Figure 6: Ethical Exploitation Demonstration

14. Financial Credential Monitoring

A financial organization wants to monitor dark web discussions related to stolen customer banking credentials. Students must identify possible threats and recommend preventive actions.

Tools: IntelligenceX, Threat Intelligence Platforms, Tor Browser

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open IntelligenceX

Open:

<https://intelx.io>

Step 4 — Search Banking Keywords

Search:

bank credentials

OR

bank login leak

Observe:

- Credential discussions
- Leak references
- Threat indicators

 SS2 → IntelligenceX search results

Step 5 — Open Threat Intelligence Platform

Open:

<https://www.virustotal.com/>

OR other threat intel platform.

Search:

bank phishing

Observe:

- Threat reports
- Malicious indicators
- Related infrastructure

 SS3 → Threat intelligence platform results

Step 6 — Analyze Threat Indicators

Identify:

- Credential leaks
- Banking phishing indicators
- Suspicious discussions
- Risk severity

15. Legal and Ethical Investigation Review

A cyber investigation team plans to monitor darknet activities. Students must identify ethical limitations, legal boundaries, and privacy concerns during investigations.

Tools: Cyber Law References, Academic Journals

Step 1 — Open Cyber Law Reference

Open:

<https://cybercrime.gov.in/>

Observe:

- Cyber laws
- Legal investigation guidelines
- Privacy regulations

 SS1 → Cyber law reference website

Step 2 — Open Academic Journal Source

Open:

<https://scholar.google.com/>

Search:

dark web investigation ethics

OR

cyber investigation privacy concerns

Observe:

- Research papers
- Ethical investigation discussions
- Privacy concerns

 SS2 → Academic journal search results

Step 3 — Identify Ethical Limitations

Observe and note:

- Privacy protection

- No unauthorized access
 - No interaction with illegal content
 - Passive investigation only
-

Step 4 — Identify Legal Boundaries

Write:

- Follow cyber laws
 - Avoid illegal monitoring
 - Do not access private systems
 - Respect jurisdiction regulations
-

Step 5 — Identify Privacy Concerns

Observe:

- Exposure of personal data
 - Monitoring limitations
 - Data handling ethics
 - User anonymity concerns
-

Output

Ethical limitations, privacy concerns, and legal investigation boundaries related to darknet monitoring were successfully identified and analyzed.

Result

Legal and ethical investigation review successfully demonstrated the importance of lawful and privacy-respecting cyber investigations.

Quick Memory

Open Cyber Law Site → Search Academic Journals → Identify Ethics → Analyze

16. Dark Web Drug Marketplace Observation

A cybercrime analyst receives information about a suspected drug marketplace hosted on Tor. Students must study the site structure and identify operational features without engaging in illegal activities.

Tools: Tor Browser, Onion Directories, Screenshot Tools

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open Safe Onion Directory

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Search:

marketplace

Observe:

- Marketplace-related pages
- Categories
- Navigation structure

DO NOT interact.

 SS2 → Onion directory/search results

Step 4 — Observe Operational Features

Identify:

- Product categories
- Vendor sections
- Payment methods
- Login/register portals
- Escrow/payment references

Write observations only.

Step 5 — Take Screenshot of Structure

Take screenshot showing:

- Homepage structure
- Navigation/menu
- Marketplace layout

🚩 SS3 → Marketplace structure observation

Step 6 — Document Findings

Write:

The observed marketplace structure included vendor listings, product categories, cryptocurrency payment references, and anonymous communication features.

Output

Operational features and structural elements of a suspected darknet marketplace were successfully observed without interacting with illegal content.

Result

Dark web marketplace observation was successfully performed ethically using Tor Browser and publicly accessible onion directories.

Quick Memory

Open Tor → Verify Connection → Open Onion Directory → Search Marketplac

17. Fake Cryptocurrency Giveaway Scam Analysis

A fake cryptocurrency investment scheme is promoted through darknet forums. Students must identify indicators of fraud and track connected wallet addresses.

Tools: Blockchain Explorer, IntelligenceX

Step 1 — Open Blockchain Explorer

Open:

<https://www.blockchain.com/explorer>

Step 2 — Search Sample Bitcoin Wallet

Search:

1BoatSLRHtKNgkdXEeobR76b53LETtpyT

Observe:

- Transactions

- Wallet activity
- Incoming/outgoing BTC
- Balance

 SS1 → Wallet transaction overview

Step 3 — Analyze Wallet Activity

Observe:

- Multiple incoming payments
 - Suspicious transfer patterns
 - Frequent small transactions
 - Wallet movement behavior
-

Step 4 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

crypto giveaway scam

OR

bitcoin scam

Observe:

- Scam discussions
- Related domains
- Fraud indicators
- Wallet references

 SS2 → IntelligenceX search results

Step 5 — Identify Fraud Indicators

Observe:

- Unrealistic investment promises
 - Anonymous payment requests
 - Repeated wallet usage
 - Suspicious communication patterns
-

Step 6 — Correlate Wallet Information

Compare:

- Wallet activity
- Scam references
- Related discussions

Document observations.

Output

Fraud indicators and suspicious cryptocurrency wallet activity related to fake giveaway scams were successfully identified and analyzed.

Result

Fake cryptocurrency giveaway scam analysis was successfully performed using blockchain analysis and OSINT investigation tools.

Quick Memory

Open Blockchain Explorer → Search Wallet → Analyze Transactions → Open In

18. Botnet Command-and-Control Investigation

A suspected botnet command server is discussed on underground forums. Students must gather OSINT related to domains, IPs, and communication channels.

Tools: Shodan, SpiderFoot, Maltego CE

Step 1 — Open Shodan

Open:

<https://www.shodan.io/>

Search:

apache

OR

nginx

Observe:

- IP addresses
- Open ports
- Hosting details
- Services

 SS1 → Shodan search results

Step 2 — Start SpiderFoot

Open terminal:

```
spiderfoot -l 127.0.0.1:5001
```

Press Enter.

Step 3 — Open SpiderFoot Dashboard

Open browser:

```
http://127.0.0.1:5001
```

Step 4 — Create New Scan

Scan Name:

BotnetScan

Scan Target:

example.com

Click:

Run Scan

Observe:

- Domains
- IP addresses
- Emails
- Related infrastructure

 SS2 → SpiderFoot scan dashboard

Step 5 — Open Maltego CE

Open terminal:

```
maltego
```

Wait for workspace.

 SS3 → Maltego workspace

Step 6 — Create Relationship Graph

Add entities:

- Domains
- IP addresses

- Communication servers
- Hosting providers

Analyze:

- Infrastructure links
 - Related entities
 - Communication patterns
-

Output

OSINT related to suspected botnet command-and-control infrastructure was successfully gathered and correlated.

Result

Botnet command-and-control investigation successfully identified domain

19. Anonymous Email Service Investigation

A suspicious anonymous email provider hosted on the Dark Web is allegedly being used for cybercrime communication. Students must analyze its structure and associated risks.

Tools: Tor Browser, Recon-ng

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Search:

anonymous email service

Observe:

- Email service references
- Login portals
- Communication features
- Site structure

DO NOT login/register.

 SS2 → Onion search results

Step 4 — Open Recon-ng

Open terminal:

recon-ng

Wait for:

[recon-ng][default] >

 SS3 → Recon-ng console

Step 5 — Optional Marketplace Search

Inside Recon-ng:

marketplace search

Observe available modules.

Step 6 — Analyze Risks

Identify:

- Anonymous communication risks
- Possible cybercrime usage
- Hidden registration details
- Lack of accountability

Document observations.

Output

The anonymous email service structure and associated cybersecurity risks were successfully analyzed using passive OSINT methods.

Result

Anonymous email service investigation successfully identified structura

20. Dark Web Identity Theft Investigation

A victim reports identity theft after personal details appeared online. Students must investigate possible data leak sources and related discussions.

Tools: IntelligenceX, DeHashed, OSINT Framework

Step 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

victim@example.com

OR

identity leak

Observe:

- Leak references
- Public exposure
- Related discussions

 SS1 → IntelligenceX search results

Step 2 — Open DeHashed

Open:

<https://dehashed.com/>

Search:

victim@example.com

Observe:

- Exposed credentials
- Email leaks
- Breach information

 SS2 → DeHashed results

Step 3 — Open OSINT Framework

Open:

<https://osintframework.com/>

Observe:

- Identity investigation categories
- Email investigation resources
- Breach analysis tools

 SS3 → OSINT Framework page

Step 4 — Analyze Possible Leak Sources

Identify:

- Breached websites
 - Credential leaks
 - Public exposure indicators
 - Related discussions
-

Step 5 — Document Findings

Write:

Possible data exposure sources and leaked identity-related information were identified through publicly available OSINT resources.

Output

Potential identity theft indicators and data leak sources were successfully in

21. Exploit Database Verification

A security researcher discovers a newly published exploit on a dark web forum. Students must safely test whether the exploit works in a lab environment.

Tools: Metasploit Framework, Kali Linux, VirtualBox

Step 1 — Start Virtual Machines

Open:

VirtualBox

Start:

- Kali Linux VM
- Vulnerable target VM (example: Metasploitable)

 SS1 → Running virtual machines

Step 2 — Identify Target IP

In vulnerable VM terminal:

```
ifconfig
```

OR

```
ip a
```

Note target IP.

Example:

```
192.168.56.101
```

🚩 SS2 → Target IP address

Step 3 — Open Metasploit

In Kali terminal:

```
msfconsole
```

Wait for:

```
msf6 >
```

🚩 SS3 → Metasploit console

Step 4 — Search Exploit

Command:

```
search vsftpd
```

Observe available exploit modules.

Step 5 — Select Exploit Module

Command:

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Step 6 — Configure Target

Set target IP:

```
set RHOSTS 192.168.56.101
```

Step 7 — Run Exploit Verification

Command:

```
run
```

Observe:

- Vulnerability verification

- Session opened
- Exploit success/failure

 SS4 → Exploit verification result

Output

The exploit was successfully tested in an isolated laboratory environment using controlled ethical procedures.

Result

Exploit database verification successfully demonstrated exploit validation using Metasploit Framework in a safe lab setup.

22. Cryptocurrency Mixing Service Analysis

Investigators suspect criminals are using cryptocurrency mixing services to hide transaction trails. Students must study transaction patterns and anonymity techniques.

Tools: Blockchain Explorer, BTC Explorer

Step 1 — Open Blockchain Explorer

Open:

<https://www.blockchain.com/explorer>

Step 2 — Search Sample Bitcoin Wallet

Search:

1BoatSLRHtKNgkdXEobR76b53LETtpyT

Observe:

- Incoming transactions
- Outgoing transactions
- Wallet balance
- Transaction count

 SS1 → Blockchain wallet overview

Step 3 — Analyze Transaction Patterns

Observe:

- Multiple transfers
- Frequent wallet hops

- Small distributed transactions
- Rapid transaction movement

These may indicate:

mixing/anonymity behavior

Step 4 — Open BTC Explorer

Open:

<https://blockchair.com/bitcoin>

Search same wallet:

1BoatSLRHtKNngdXEeobR76b53LETtpyT

Observe:

- Transaction graph
- Connected wallets
- Wallet behavior
- Address relationships

 SS2 → Blockchair wallet analysis

Step 5 — Identify Anonymity Techniques

Observe:

- Wallet chain movement
- Multiple output addresses
- Transaction obfuscation
- Repeated transfers

Document findings.

Output

Transaction patterns and anonymity-related behaviors associated with cryptocurrency mixing techniques were successfully analyzed.

Result

Cryptocurrency mixing service analysis successfully demonstrated how blockchain investigations can identify suspicious transaction patterns.

A government organization receives threat alerts from multiple intelligence feeds.

Students must correlate information from darknet forums, leaks, and public sources.

Tools: Maltego CE, SpiderFoot, IntelligenceX

Step 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

government cyber threat

OR

data leak

Observe:

- Leak references
- Threat discussions
- Related intelligence

 SS1 → IntelligenceX search results

Step 2 — Start SpiderFoot

Open terminal:

```
spiderfoot -l 127.0.0.1:5001
```

Press Enter.

Step 3 — Open SpiderFoot Dashboard

Open browser:

<http://127.0.0.1:5001>

Step 4 — Create New Scan

Scan Name:

ThreatIntelScan

Scan Target:

gov.in

Click:

Run Scan

Observe:

- Domains

- Emails
- IPs
- Related intelligence

🚩 SS2 → SpiderFoot scan dashboard

Step 5 — Open Maltego CE

Open terminal:

maltego

Wait for workspace.

🚩 SS3 → Maltego workspace

Step 6 — Create Correlation Graph

Add entities:

- Domains
- Threat actors
- IP addresses
- Leaks
- Forums

Analyze:

- Relationships
 - Connected infrastructure
 - Threat patterns
-

Output

Threat intelligence from multiple sources was successfully correlated using OSINT and intelligence analysis tools.

Result

Threat intelligence correlation successfully combined data from darknet

24. Fake Login Page Investigation

A cloned university login page is circulating on hidden forums. Students must identify hosting details and indicators of phishing attacks.

Tools: URLScan, WHOIS Lookup, Tor Browser

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open URLScan

Open:

<https://urlscan.io/>

Search sample phishing URL:

<http://fake-university-login.com>

Observe:

- Hosting IP
- Domain behavior
- Related resources
- Suspicious indicators

 SS2 → URLScan analysis results

Step 4 — Open WHOIS Lookup

Open:

<https://whois.domaintools.com/>

Search:

fake-university-login.com

Observe:

- Registrar
- Creation date
- Hosting provider
- Domain details

Step 5 — Identify Phishing Indicators

Observe:

- Recently registered domain
- Similar login page names
- Suspicious hosting
- Fake branding indicators

Document findings.

Output

Hosting details and phishing indicators related to the fake university login page were successfully identified and analyzed.

Result

Fake login page investigation successfully demonstrated phishing analysis using URLScan, WHOIS lookup, and Tor Browser.

Quick Memory

Open Tor → Verify Connection → Analyze URL with URLScan → Perform WHOIS Lookup

25. Onion Website Risk Assessment

A newly discovered onion website claims to sell hacking tools. Students must perform a risk assessment and classify the type of cyber threat involved.

Tools: Tor Browser, Maltego CE

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Search:

hacking tools

Observe:

- Website categories
- Threat-related content
- Communication methods
- Payment references

DO NOT interact.

 SS2 → Onion search results

Step 4 — Identify Threat Indicators

Observe:

- Malware/tool references
- Anonymous payment methods
- Illegal service indicators
- Cybercrime-related terminology

Classify risk:

High Risk Cyber Threat

Step 5 — Open Maltego CE

Open terminal:

maltego

Wait for workspace.

 SS3 → Maltego workspace

Step 6 — Create Risk Analysis Graph

Add entities:

- Onion website
- Threat category
- Payment methods

- Communication channels

Analyze relationships.

Output

The onion website was assessed and classified as a high-risk cyber threat based on observed operational and cybercrime-related indicators.

Result

Onion website risk assessment successfully demonstrated threat classification and passive analysis using Tor Browser and Maltego CE.

26. Data Leak Trend Analysis

A cyber intelligence team wants to study trends in leaked databases shared on the Dark Web over the last six months.

Tools: IntelligenceX, Spreadsheet Tools, OSINT Framework

Step 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

database leak

OR

data breach

Observe:

- Leak references
- Dates
- Breach categories
- Exposed data types

 SS1 → IntelligenceX search results

Step 2 — Collect Sample Observations

Note:

- Leak names
- Dates
- Type of leaked data

- Source/platform

Example:

Email leaks, credential leaks, database dumps

Step 3 — Open Spreadsheet Tool

Open:

- Excel
OR
- Google Sheets

Create columns:

| Month | Leak Type | Data Exposed | Source |

Step 4 — Enter Sample Data

Example:

Month	Leak Type	Data Exposed	Source
January	Credential Leak	Emails & Passwords	Forum
February	Database Leak	User Records	Breach Site
March	Banking Leak	Financial Data	Dark Web

🇮🇸 SS2 → Spreadsheet trend analysis

Step 5 — Open OSINT Framework

Open:

<https://osintframework.com/>

Observe:

- Breach investigation resources
- Leak analysis categories
- OSINT investigation tools

🇮🇸 SS3 → OSINT Framework page

Step 6 — Analyze Trends

Identify:

- Increasing breach frequency
- Common leak types
- Frequently exposed data

- Threat patterns

Document findings.

Output

Data leak trends and common breach patterns over recent months were successfully analyzed using OSINT resources and spreadsheet-based analysis.

Result

Data leak trend analysis successfully demonstrated how cyber intelligence teams

27. Social Engineering Campaign Investigation

A phishing campaign targeting employees is being discussed on underground forums.

Students must identify social engineering techniques used by attackers.

Tools: Tor Browser, SpiderFoot, IntelligenceX

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

🇮🇸 SS1 → Tor verification page

Step 3 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

employee phishing

OR

social engineering

Observe:

- Phishing discussions

- Credential theft references
- Attack techniques

 SS2 → IntelligenceX search results

Step 4 — Start SpiderFoot

Open terminal:

```
spiderfoot -l 127.0.0.1:5001
```

Press Enter.

Step 5 — Open SpiderFoot Dashboard

Open browser:

<http://127.0.0.1:5001>

Step 6 — Create New Scan

Scan Name:

PhishingScan

Scan Target:

company.com

Click:

Run Scan

Observe:

- Domains
- Emails
- Related infrastructure
- Threat indicators

 SS3 → SpiderFoot scan dashboard

Step 7 — Identify Social Engineering Techniques

Observe:

- Fake login pages
- Credential harvesting
- Urgent messages
- Impersonation tactics
- Psychological manipulation

Document findings.

Output

28. Secure File Sharing Service Investigation

A hidden file-sharing service is suspected of distributing malicious files. Students must examine file distribution methods and possible security risks.

Tools: Tor Browser, VirusTotal, REMnux

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Search:

file sharing

Observe:

- File-sharing references
- Download sections
- Anonymous distribution methods

DO NOT download files.

 SS2 → Onion search results

Step 4 — Open VirusTotal

Open:

<https://www.virustotal.com/>

Search:

malware file

Observe:

- Threat detections
- File behavior
- Malicious indicators

 SS3 → VirusTotal analysis results

Step 5 — Start REMnux VM

Open:

VirtualBox

Start:

REMnux VM

 SS4 → REMnux desktop

Step 6 — Analyze Security Risks

Identify:

- Malware distribution methods
- Anonymous hosting risks
- Suspicious file behavior
- Possible ransomware indicators

Document findings.

29. Dark Web Threat Report Preparation

A cyber security company wants a summary report about current dark web cyber threats targeting educational institutions.

Tools: IntelligenceX, Threat Intelligence Platforms, Spreadsheet Tools

Step 1 — Open IntelligenceX

Open:

<https://intelx.io>

Search:

education cyber attack

OR

university data breach

Observe:

- Threat discussions
- Breach references
- Educational institution targeting

 SS1 → IntelligenceX search results

Step 2 — Open Threat Intelligence Platform

Open:

<https://www.virustotal.com/>

Search:

education phishing

Observe:

- Phishing indicators
- Malicious domains
- Threat detections

 SS2 → Threat intelligence results

Step 3 — Open Spreadsheet Tool

Open:

- Excel
OR
- Google Sheets

Create columns:

| Threat Type | Target | Impact | Recommendation |

Step 4 — Enter Sample Threat Data

Example:

Threat Type	Target	Impact	Recommendation
Phishing	Students	Credential Theft	Enable MFA
Ransomware	University Servers	Data Encryption	Backup Systems
Data Breach	Student Records	Privacy Exposure	Reset Passwords

 SS3 → Threat report spreadsheet

Step 5 — Analyze Threat Trends

Identify:

- Common attack types
- Educational targets
- Threat severity
- Preventive measures

Document findings.

30. Anonymous Chat Platform Investigation

Law enforcement suspects cybercriminals are coordinating attacks using anonymous dark web chat platforms. Students must analyze communication channels and identify possible risks.

Tools: Tor Browser, Maltego CE, OSINT Framework

Step 1 — Open Tor Browser

Start:

Tor Browser

Step 2 — Verify Tor Connection

Open:

<https://check.torproject.org>

Expected:

Congratulations. This browser is configured to use Tor.

 SS1 → Tor verification page

Step 3 — Open Safe Onion Search Engine

Open:

<http://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion>

Search:

anonymous chat

Observe:

- Chat platform references

- Communication channels
- Anonymous messaging features

DO NOT interact.

 SS2 → Onion search results

Step 4 — Open OSINT Framework

Open:

<https://osintframework.com/>

Observe:

- Communication investigation tools
- OSINT analysis categories
- Threat intelligence resources

 SS3 → OSINT Framework page

Step 5 — Open Maltego CE

Open terminal:

maltego

Wait for workspace.

 SS4 → Maltego workspace

Step 6 — Create Communication Analysis Graph

Add entities:

- Chat platforms
- Usernames
- Communication channels
- Threat categories

Analyze:

- Relationships
 - Communication patterns
 - Possible cyber risks
-

Step 7 — Identify Risks

Observe:

- Anonymous coordination

- Attack planning indicators
- Hidden identities
- Lack of traceability

Document findings.

Output

Anonymous communication channels and associated cybercrime risks were successfully analyzed using passive OSINT investigation techniques.

Result

Anonymous chat platform investigation successfully demonstrated communication analysis and cyber risk identification using Tor Browser, Maltego CE, and OSINT Framework.

Quick Memory